

VM-Series on Azure — Deployment Questionnaire

Information required to build and customize your Terraform deployment configuration

This questionnaire collects the information needed to configure a VM-Series transit VNet deployment on Azure using the official [Palo Alto Networks Terraform modules](#). Complete each section and return this document so we can prepare your customized `terraform.tfvars` and deployment package.

ABOUT EXISTING INFRASTRUCTURE

The Terraform modules are designed to deploy the complete transit VNet — VNet, subnets, NSGs, route tables, load balancers, and VM-Series firewalls — as a single managed unit. This is the **preferred approach** because the firewall deployment has many tightly coupled moving parts: subnet-to-NSG bindings, blackhole UDRs between zones, load balancer backend pool associations, and availability zone placement all need to be coordinated precisely.

That said, **we can modify the Terraform to work with existing infrastructure** — for example, peering into a VNet you've already provisioned, reusing existing NSGs or route tables, or attaching to load balancers managed outside of Terraform. This is common in enterprises where networking teams provision VNets independently, or where the transit VNet was created through a landing zone accelerator.

Section 2 below asks you to describe what infrastructure is already in place (or will be built by another team) so we know what Terraform needs to create vs. reference. If you answer "greenfield" (Terraform builds everything), no extra integration work is needed.

1 Azure Subscription & Identity

Azure Account

Azure Subscription ID

GUID format: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx

Azure Tenant ID

Your Azure AD / Entra ID tenant

Target Azure Region

e.g., East US, West Europe, Southeast Asia

Resource Group Name

For the transit VNet and firewalls

Name Prefix

Prepended to all resource names (e.g., prod- , customer-)

Terraform Identity

How will Terraform authenticate to Azure?

Authentication method

- ☐ Logged-in user via `az login` (dev/test)
- ☐ Service Principal (App Registration) — preferred for CI/CD
- ☐ Managed Identity
- ☐ Azure Cloud Shell (portal)

Service Principal App ID

If using SP auth (leave blank if not)

Resource Tags

Tags applied to all resources. List any required tags (e.g., cost center, environment, owner, project).

TAG KEY	TAG VALUE

2 Existing Infrastructure Assessment

WHY THIS MATTERS

The VM-Series transit VNet deployment creates a VNet with specialized subnets (management, public/untrust, private/trust), NSGs with inter-zone blackhole rules, route tables with specific UDR chains, and load balancers with health probes — all tightly interdependent. Letting Terraform create and manage these as a unit is the cleanest path. However, if your organization has already provisioned some of these components (or a different team will), we need to know so we can wire Terraform to reference them via `data` sources instead of creating them.

Transit VNet

Transit VNet status

-  **Greenfield** — Terraform should create the transit VNet, subnets, NSGs, and route tables from scratch (recommended)
-  **VNet exists** — The transit VNet is already provisioned but the firewall subnets need to be created within it
-  **VNet + subnets exist** — Both the VNet and the required subnets (mgmt, public, private) already exist
-  **Full brownfield** — VNet, subnets, NSGs, route tables, and/or load balancers already exist and should be reused

If the VNet exists, provide:

VNet name and resource group

Existing subnet names

If subnets already exist, list the names for
mgmt, public, private

Network Security Groups & Route Tables

NSGs managed by

- ☐ Terraform (this deployment) — recommended
- ☐ Existing NSGs managed by another team / tool — provide names below
- ☐ Azure Policy auto-assigns NSGs

Route tables managed by

- ☐ Terraform (this deployment) — recommended
- ☐ Existing route tables managed by another team / tool

Load Balancers

Load balancers managed by

- ☐ Terraform (this deployment) — recommended
- ☐ Existing load balancers — provide names and resource group below

Existing LB details

If LBs exist, provide public LB name and private (internal) LB name

Spoke VNets

Spoke VNets to peer with the transit

- ☐ No spoke VNets yet — we'll add them later
- ☐ Existing spoke VNets need peering — list below
- ☐ Terraform should create test/sample spoke VNets

SPOKE VNET NAME	RESOURCE GROUP	ADDRESS SPACE	PURPOSE

Panorama VNet Peering

Panorama runs in Azure?

- ☐ Yes — Panorama is in an Azure VNet and we need VNet peering
- ☐ No — Panorama is on-prem or accessed via VPN/ExpressRoute
- ☐ Not using Panorama (SCM only)

Panorama VNet name

If Panorama is in Azure

Panorama resource group

If Panorama is in Azure

Terraform State Backend

Terraform state backend

- ☐ We need Terraform to create the state backend (Storage Account)
- ☐ We already have a state backend — provide details below
- ☐ Using Terraform Cloud / HCP Terraform
- ☐ Local state only (not recommended for production)

State Storage Account name

If the backend already exists

State container name

Default: `tfstate`

State key path

Default: `pan-vmseries/azure/terraform.tfstate`

3 Architecture Model

COMMON MODEL VS. DEDICATED MODEL

- **Common Model** — All firewall instances handle all traffic directions (inbound, outbound, east-west). Simpler to manage, fewer VMs, good for most deployments. Typically 2 firewalls across availability zones.
- **Dedicated Model** — Separate firewall pairs for inbound vs. outbound/east-west traffic. Higher throughput ceiling, independent scaling per direction, but more VMs (typically 4+). Choose this for high-throughput environments or when inbound and outbound security policies diverge significantly.

Architecture model

- ❏ **Common Model** — 2 firewalls handle all traffic directions
- ❏ **Dedicated Model** — Separate inbound pair + outbound/east-west pair
- ❏ Not sure — need guidance based on throughput and use case

4 Network & IP Planning

SUBNET SIZING

The transit VNet needs at least 3 subnets (management, public/untrust, private/trust), each with enough IPs for the firewall count. A `/28` (14 usable IPs) per subnet is typical. If using Application Gateway (Common Model), a 4th `/28` subnet is needed. The default example uses a `/25` VNet CIDR to accommodate all subnets.

Transit VNet Address Space

Transit VNet CIDR

e.g., `10.0.0.0/25` — must be unique in your environment

Subnet CIDRs

Provide the CIDR for each subnet within the transit VNet, or leave blank to use defaults based on your VNet CIDR.

SUBNET	PURPOSE	DEFAULT (FROM /25)	YOUR CIDR
Management	Firewall mgmt plane (eth0)	x.x.x.0/28	
Public (Untrust)	Firewall untrust interface (eth1)	x.x.x.16/28	
Private (Trust)	Firewall trust interface (eth2)	x.x.x.32/28	
AppGW (<i>Common only</i>)	Application Gateway subnet	x.x.x.48/28	

Internal Load Balancer IP

Private LB frontend IP

Must be within the private/trust subnet.

This is the next-hop for all spoke UDRs.

Default: last usable IP in the private
subnet.

RFC 1918 / Internal Supernet

Internal CIDR supernet

Used in bootstrap templates for internal routing. Default: 10.0.0.0/8

5 Management Access

Management NSG — Allowed Source IPs

These IP addresses / CIDRs will be allowed inbound to the management subnet on ports 22 (SSH) and 443 (HTTPS). List all admin IPs, jump-box ranges, VPN ranges, and Panorama IPs that need to reach the firewall management plane.

IP / CIDR	DESCRIPTION

6 Inbound Applications (Public Load Balancer)

Each inbound application gets a public IP on the public load balancer with forwarding rules to the firewall's untrust interface. The firewalls inspect the traffic and DNAT it to the backend application in a spoke VNet.

Public LB Frontend Applications

List each application or service that needs an inbound public IP through the firewalls. Each entry becomes a frontend IP + LB rule on the public load balancer.

APP NAME	PROTOCOL	PORT(S)	BACKEND DESTINATION (SPOKE IP)	BRING EXISTING PUBLIC IP?

Source IP allowlist for LB

Public IPs/CIDRs allowed to reach the LB frontend (or * for any)

Application Gateway (Common Model Only)

Use Application Gateway?

AppGW provides L7 features (SSL offload, WAF, path-based routing) in front of the firewalls

- ☐ Yes — deploy Application Gateway with the firewalls as backend
- ☐ No — public load balancer only
- ☐ Not sure

7 VM-Series Configuration

Firewall Sizing & Version

PAN-OS Version

Default: 11.1.607 . SCM requires 11.0+.

Azure VM Size

Default: Standard_DS3_v2 (4 vCPU, 14 GB RAM). Larger for higher throughput.

Marketplace offer plan



byol — Bring Your Own License (Software NGFW Credits)



bundle1 — PAYG Bundle 1



bundle2 — PAYG Bundle 2

Firewall Count & Availability Zones

Common Model (if selected in Section 3)

FIREWALL	NAME	AVAILABILITY ZONE	PUBLIC IP ON MGMT?
FW-1			
FW-2			

Dedicated Model (if selected in Section 3)

ROLE	FIREWALL	NAME	AVAILABILITY ZONE	PUBLIC IP ON MGMT?
Inbound	FW-IN-1			
	FW-IN-2			
OBEW	FW-OBEW-1			
	FW-OBEW-2			

NOTE

Spreading firewalls across availability zones (e.g., Zone 1 and Zone 2) provides high availability. Azure guarantees 99.99% SLA for VMs distributed across zones.

8 Management Platform & Bootstrap

When firewalls boot, they use **bootstrap options** to auto-register with a management platform. Pick one path and fill in the corresponding values.

Management Platform

Management platform

- ☐ **Panorama** — on-prem or Azure-hosted Panorama with SW Firewall License Plugin
- ☐ **Strata Cloud Manager (SCM)** — SaaS management plane (requires PAN-OS 11.0+)
- ☐ **None (standalone)** — basic bootstrap only, no central management

8A. Panorama Bootstrap Values

Complete this section if using Panorama. These values come from the SW Firewall License Plugin configuration on Panorama.

Panorama IP address

Primary Panorama server IP reachable
from the firewall mgmt interface

Template Stack name

e.g., TPL-AZ-COMMON-STACK

Device Group name

e.g., DG-AZ-COMMON

Auth Key

Generated from the License Manager
bootstrap definition on Panorama

Secondary Panorama IP

Optional — for HA Panorama pairs

Dedicated Model Only — Separate Bootstrap per Tier

If using the Dedicated Model, inbound and OBEW firewalls typically have separate Template Stacks and Device Groups.

VALUE	INBOUND TIER	OBEW TIER
Template Stack name		
Device Group name		
Auth Key		

8B. SCM Bootstrap Values

Complete this section if using Strata Cloud Manager. These values come from the CSP and SCM consoles.

SCM Folder name

The folder in SCM where the firewalls will be managed (maps to `dgname`)

SCM Device Label

Label for grouping devices in SCM (maps to `tplname`)

Auto-Registration PIN ID

UUID from Customer Support Portal

Auto-Registration PIN Value

Hex string from Customer Support Portal

License Auth Code

Deployment profile auth code (format: `D1234567`)

9 Licensing & Credits

Licensing model

- ☐ Software NGFW Credits (BYOL with deployment profiles)
- ☐ PAYG (pay-as-you-go via Azure Marketplace)
- ☐ ELA / existing entitlement

Credit pool activated?

- ☐ Yes
- ☐ No — not yet activated
- ☐ N/A (PAYG)

Deployment profile auth code

Format: D1234567 . One per distinct vCPU + subscription bundle.

Auto-Registration PIN ID

UUID — required for device certificate enrollment

Auto-Registration PIN Value

Hex string

DEDICATED MODEL — MULTIPLE PROFILES

If your inbound and OBEW tiers use different vCPU counts or subscription bundles, you'll need separate deployment profiles (and therefore separate auth codes) for each tier. List both if applicable.

Dedicated Model — Per-Tier Auth Codes (if different)

TIER	AUTH CODE	VCPU COUNT	SUBSCRIPTIONS
Inbound			
OBEW			

10 Monitoring & Observability

Azure NGFW Metrics

Publishes firewall metrics to Azure Monitor for dashboards and alerts



Yes — enable Azure NGFW Metrics



No — skip for now

Azure Metrics workspace name

Name for the metrics resource (default: metrics)

11 Additional Considerations

Connectivity

On-prem connectivity

- ☐ ExpressRoute
- ☐ Site-to-Site VPN
- ☐ Azure Virtual WAN
- ☐ None — cloud-only deployment

On-prem network CIDRs

For routing and security policy planning

CI/CD & Automation

Deployment method

- ☐ Manual terraform apply from local machine / Cloud Shell
- ☐ GitHub Actions
- ☐ Azure DevOps Pipelines
- ☐ GitLab CI
- ☐ Terraform Cloud / HCP Terraform

Timeline & Environment

Target deployment date

Environment type



Production



Pre-production / Staging



Dev / Lab / POC

**Change management
requirements**

CAB approval, maintenance windows, etc.

Open Questions / Special Requirements

Anything else we should know about your environment, constraints, or requirements?

NEXT STEPS

Complete this questionnaire and return it to your Palo Alto Networks SE. We will use your answers to:

1. Build a customized `terraform.tfvars` matching your environment
2. Identify any brownfield integration work needed (existing VNets, LBs, NSGs)
3. Configure the appropriate Panorama or SCM bootstrap parameters
4. Prepare a deployment walkthrough tailored to your architecture model

If you're unsure about any field, leave it blank and note your question in Section 11. We'll work through it together.